

Measurement & Geometry

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1 Units

1. Litres (L) used for liquids
2. Metres (m) used for length
3. Grams (g) used for mass

2 Prefixes

1. milli
2. centi
3. kilo
4. mega

Prefixes are words we put in front of units (metres, litres, grams).

There are 1000 milli in one unit, (E.g: 1 gram = 1000 milligram, 1 metre = 1000 millimetre).
100 centi is in one unit, 1 kilo is 1000 units, and 1 mega is 1000000 units.

We rarely ever use centi in anything other than metres.

3 Area

Theorem 3.1 – Circle

$$\text{Area} = \pi r^2, \text{Circumference} = 2\pi r$$

Theorem 3.2 – Triangle

$$\text{Area} = \frac{1}{2}bh$$

Theorem 3.3 – Trapezium

$$\text{Area} = \frac{1}{2}(a + b)h$$

4 Measurement Conversions

Theorem 4.1 If we fill a 10cm by 10cm by 10cm box with water, we would've used 1 litre of water, and 1 litre of water will weigh exactly 1 kilogram.

Think about it, in a cubic metre box (1m by 1m by 1m), we can actually fit $100 \times 100 \times 100 = 1000000$ smaller boxes that are 1cm by 1cm by 1cm.

Now, think of a square. If we were to double the side length of a square, what happens to its area? What about if we triple it? Think about a cube also. What happens to its volume if we double, triple its side lengths?

5 Problems

Problem 5.1 How big is a $1m^3$ box, in mm^3 ?

Problem 5.2 I have 8 litres of water that I want to store in a cube box. What is the dimension of the box that I'll need?

Problem 5.3 How many millilitres are in a megalitre?

Problem 5.4 Car fuel costs \$2.70 per litre. Bob spent \$108 on fuel, and used it to travel 100km. How many km can Bob travel with 1 litre of fuel?

Problem 5.5 I have a triangle, and I will double the length of each side to get a new triangle. By what factor did the area of my triangle increase by?

Problem 5.6 A metal disc has a diameter of 12cm. A hole is cut from the centre of the disc with radius one third of the original disc. What is the area of the disc remaining?

Problem 5.7 How much string is required to tie up a parcel $2 \times 5 \times 12$ cm allowing 10% extra for the knot at the top? Write your answer in metres.

Problem 5.8 Sam has a box that is 12cm by 20cm by 7cm. He wants to pack his miniature cars in the box. They are in packets which measure 6cm by 3cm by 2cm. What is the maximum number of miniature cars that he could pack in the box?

Problem 5.9 A rectangular whiteboard is 4 times as long as it is wide. If it were 7 metres shorter and 2 metres wider, it would be a square. What is the area of the whiteboard? Express your answer in square centimetres.

Problem 5.10 Tanya can shovel $8m^3$ of dirt in 3 hours. Nikora can shovel $6m^3$ of dirt in 4 hours.

What is the total amount of dirt that they will both have shovelled if they work at the same rate for 204 hours over the next month?

Problem 5.11 A trapezium has a base that is three times longer than its parallel side. It has a height that is two-thirds the base. If the trapezium has an area of $676cm^2$, what is the length of the base?

Problem 5.12 There is a cube and a cuboid which have volumes in a ratio of 7 : 24. The cuboid's length is 4 times the length of the cube. The cuboid's width is 3 times the length of the cube. What fraction of the cube's side is the height of the cuboid?

Express your answer as a fraction in its simplest form.

Problem 5.13 A lift has a weight limit of 1500kg. If the average weight of the people in the lift is 80kg and the combined weight of all the people is 100kg over the limit, how many people are in the lift?

Problem 5.14 A cubic lamington cake, with side xcm , is covered in 1cm thick icing on all sides. Find an expression for the volume of icing.

Problem 5.15 I bend a 12cm piece of wire into a square and then into a 3, 4, 5 triangle. What is the difference in areas of these two shapes?

Problem 5.16 The four interior angles of a quadrilateral are x° , $(3x)^\circ$, $(x+20)^\circ$, $(x+40)^\circ$. Find the size in degrees of the largest angle in this quadrilateral.

Problem 5.17 27 small cubes are put together to make a bigger cube of side length 3. If all sides but the bottom are painted red, how many small cubes have only two sides painted?

Problem 5.18 A square ended cuboidal package has a volume of 300 cubic centimetres. If the dimensions are all in whole centimetres, what are the dimensions of the package that has the least surface area?

Problem 5.19 I have a length of rope that I want to use to mark off a rectangular area on the grass, I use the whole length every time. I can make many different rectangles, two I made have an area of $96cm^2$ and an area of $225cm^2$. What is the greatest rectangular area I can make with this length of rope?